MILESTONE 12

SUBMISSION CHECKLIST AND COVER SHEET

DATE:	December 1, 1997	
TO :	Dr. Linda Behrens	
FROM:	TEAM MEMBERS	WASHINGTON, ANGELA
		RAGHURAMAN, ANANTH

We are submitting the following milestone for your evaluation

Milestone 12 checklist:

[ ] We are resubmitting our System Profile from Milestone 2.

[ ] We are resubmitting our on-line inputs and outputs designed in Milestone 11.

[X] We are submitting a state transition diagram (or equivalent) that shows the relationship between display screens used in our system.

[X] We are submitting a template, on a Terminal Screen Display Layout Chart, that describes how all terminal screens are partitioned into areas for designated uses (e.g. data, messages, instruction, etc.).

[X] We are submitting Terminal Screen Display Layout Charts for each display screen in our system.

[X] We are submitting memos or attachments that describes any special terminal displays attributes to be used (such as blinking fields), message, or help instruction associated with the Terminal Screen Display Layout Charts.

[X] We are submitting sketches or sample Terminal Screen Display Layout Charts of our system's screens using realistic sample data.

[ ] We are submitting the following additional documentation:

____

We have carefully checked our terminal dialogue specification for the following:

[X] Each dialogue screen provides instructions on how to proceed, backup, and exit so that the user is always aware what to next.

[X] The screen are formatted so that the various type information, instruction, and message always appear in the same general displays area outlined by our screen template.

[X] The dialogue appearing with the body window of our screens is limited to one idea per frame.

[X] We used scrolling when the information to be displayed on the screen was continuous in nature, as in the case of most reports.

[X] All messages, instruction, or information remain on the screen long enough for users to read them.

[X] We used terminal display attributes sparingly.

[X] We simplified complex functions and reduced the amount of typing by the users by defining functions keys.

[X] We specified default values and answer for fields to be entered by the users.

[X] Our dialogue was designed to anticipate errors users might make.

[X] Our dialogue uses simple, grammatically correct sentences, and cannot be perceived by the users as funny, cute, or condescending. It does not use computer jargon abbreviations, and symbology.

[ ] Special comments to the instructor: ______________________

____

To be completed by the instructor:

I have reviewed Milestone 12. My evaluation follows:

[ ] Milestone returned ungraded because of______________________

____

[√] Milestone acceptable as submitted. Grade of 40
recorded in students' record.

[ ] Milestone acceptable as submitted; however, it needs to be corrected and resubmitted to establish a permanent grade.

[ ] Milestone **NOT** acceptable as submitted. Grade of ________
recorded in students' record.

[ ] Milestone **NOT** acceptable as submitted because __________

____

Resubmit by date of________________ to establish a permanent grade.

Note:

Please Refer to Previous Milestone for Input/Output Display Sketches

TERMINAL SCREEN DISPLAY LAYOUT FORM

☐ INPUT ______________

☐ OUTPUT ______________

APPLICATION DETAIL REPORT ON STUDENTS

SCREEN NO. ______________ SEQUENCE ______________

COLUMN: 1 – 10 | 11 – 20 | 21 – 30 | 31 – 40 | 41 – 50 | 51 – 60 | 61 – 70 | 71 – 80

ROW	
01	
02	
03	
04	WILLIAM KAHAN
05	PIN: 1227
06	SOCIAL: 241345636
07	TYPE: GRAD
08	TERM: SPR77
09	RACE: HISPANIC
10	MAJOR: COMPSCI
11	MINOR: NONE
12	SEX: M
13	DOB: 17-07-77
14	TEMP
15	ADD: 12 NORFWOOD DRIVE
16	JAMAICA NY, NY 11100
17	
18	
19	
20	
21	
22	

480

1960

TERMINAL SCREEN DISPLAY LAYOUT FORM

☒ INPUT

☐ OUTPUT

APPLICATION Access Validator

SCREEN NO. 1 SEQUENCE

COLUMN

ROW

08: ENTER PIN :

12: 1. ENTER 480 2. CANCEL 1960

TERMINAL SCREEN DISPLAY LAYOUT FORM

☐ INPUT ____________

☐ OUTPUT ____________

APPLICATION THIRD MENU

SCREEN NO. 3 SEQUENCE ____________

COLUMN

ROW	
01	
02	
03	
04	
05	
06	
07	1. EXCEPTION REPORT ON COURSES
08	2. DETAIL REPORT ON COURSES
09	3. SUMMARY REPORT ON COURSES
10	
11	1960
12	1. ENTER 480 2. CANCEL
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	

TERMINAL SCREEN DISPLAY LAYOUT FORM

☐ INPUT ____________

☐ OUTPUT ____________

APPLICATION Student PIN INPUT

SCREEN NO. ____________ SEQUENCE ____________

COLUMN 1 – 10, 11 – 20, 21 – 30, 31 – 40, 41 – 50, 51 – 60, 61 – 70, 71 – 80

ROW 01–22

03: ENTER PIN WHOSE RECORD YOU WANT TO RETRIEVE

06: ENTER PIN :

08: 0. ENTER

09: 1. CANCEL

480

1960

TERMINAL SCREEN DISPLAY LAYOUT FORM

☐ INPUT ____________

☐ OUTPUT ____________

APPLICATION SECOND MENU

SCREEN NO. 4 SEQUENCE ____________

COLUMN

ROW	
01	
02	
03	
04	
05	
06	
07	1. EXECUTION REPORT ON STUDENTS AND UPDATE
08	2. DETAIL REPORT ON STUDENTS AND UPDATE
09	3. SUMMARY REPORT ON STUDENTS AND UPDATE
10	
11	1960
12	1. ENTER 480 2. CANCEL
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	

TERMINAL SCREEN DISPLAY LAYOUT FORM

☒ INPUT

☐ OUTPUT

APPLICATION MAIN MENU

SCREEN NO. 2 SEQUENCE

COLUMN

1 – 10 | 11 – 20 | 21 – 30 | 31 – 40 | 41 – 50 | 51 – 60 | 61 – 70 | 71 – 80

ROW

01

02

03

04

05

06 1. UPDATE STUDENT RECORDS

07

08 2. UPDATE COURSE RECORDS

09

10

11 480 1960

12

13

14

15

16

17

18

19

20

21

22

TERMINAL SCREEN DISPLAY LAYOUT FORM

☐ INPUT ______________

☐ OUTPUT ______________

APPLICATION: COURSE AND SECTION NUMBER INPUT

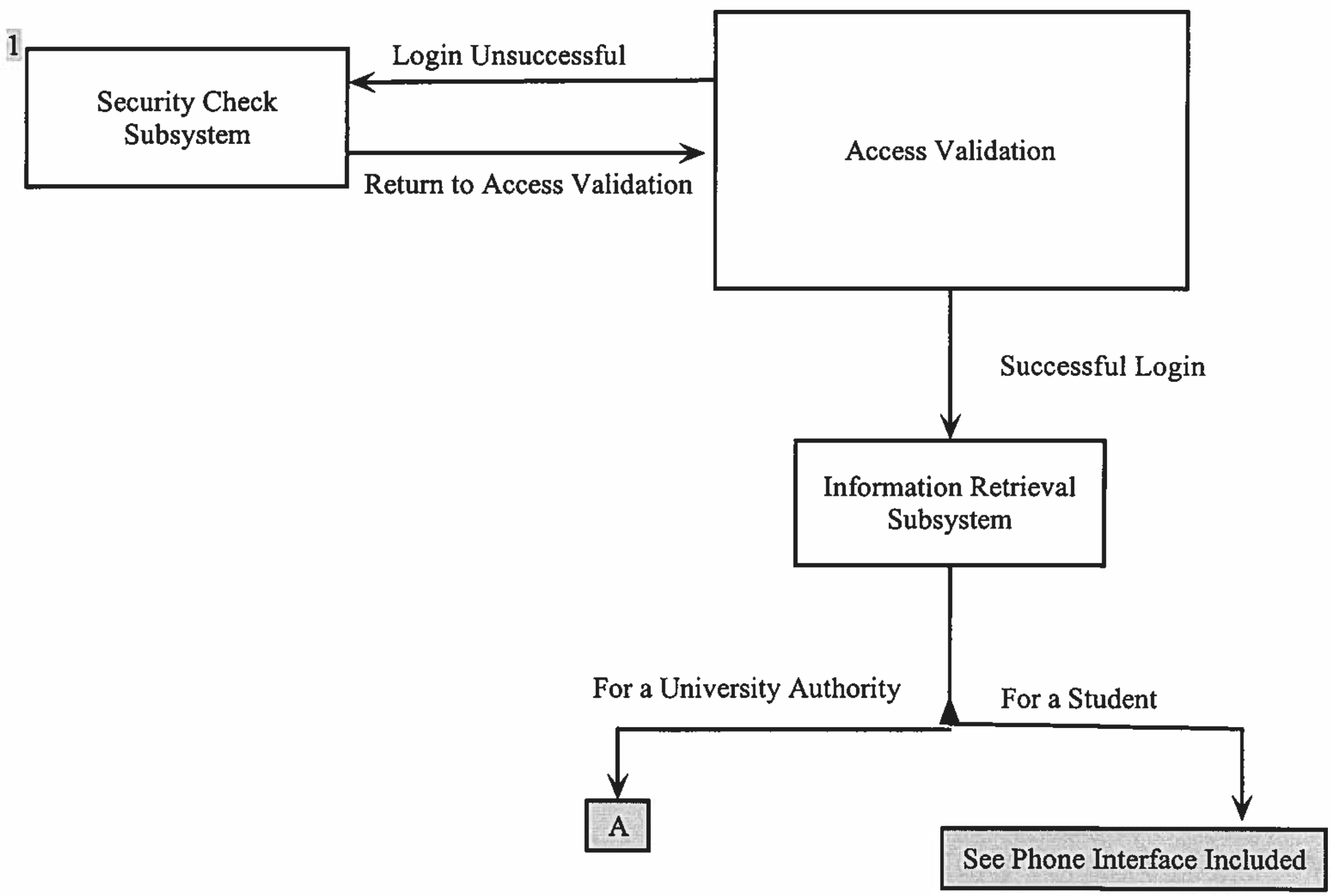
SCREEN NO. ______________ SEQUENCE ______________

COLUMN: 1 – 10, 11 – 20, 21 – 30, 31 – 40, 41 – 50, 51 – 60, 61 – 70, 71 – 80

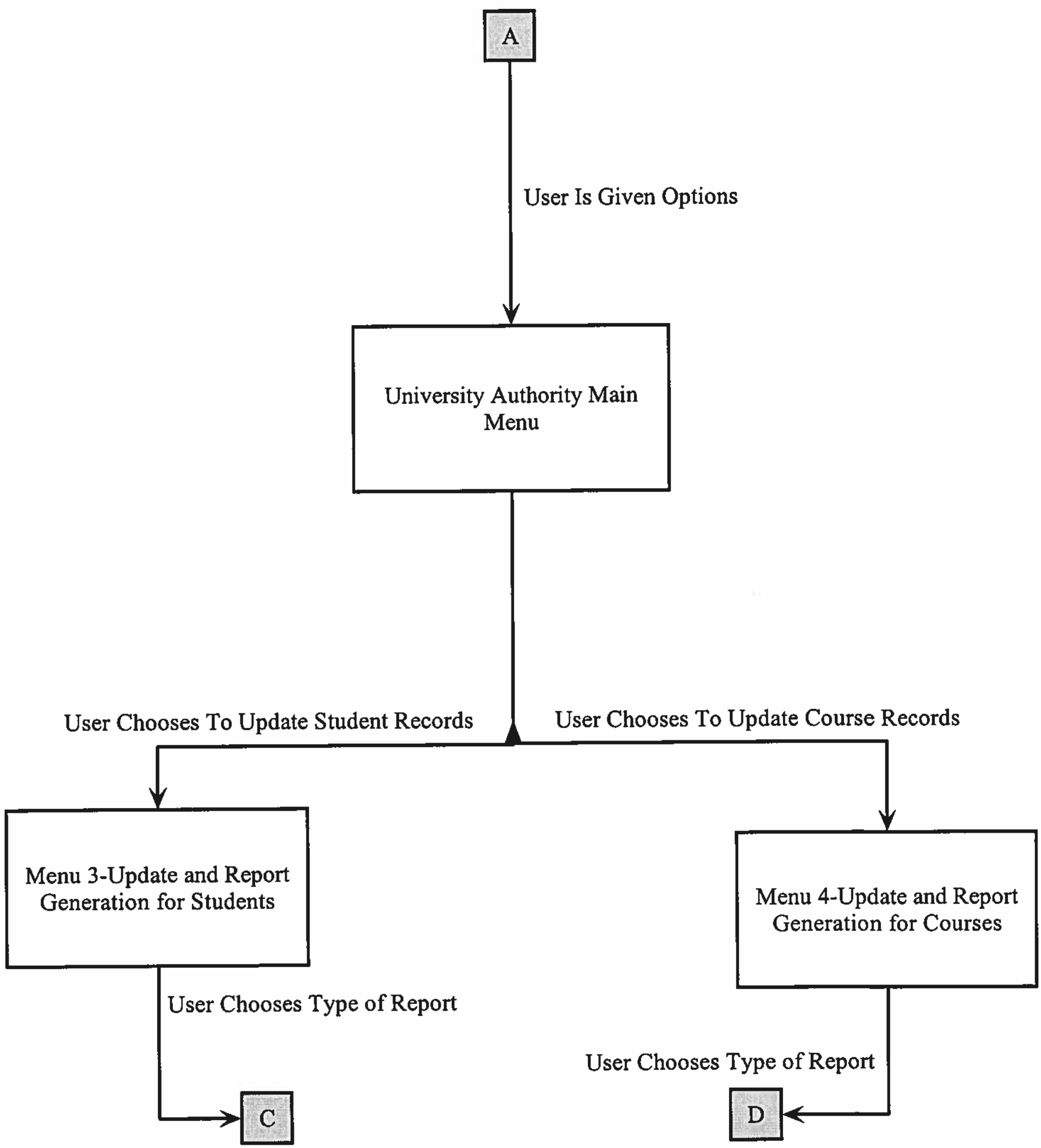
ROW: 01–22

ROW	
01	
02	
03	
04	
05	
06	ENTER COURSE, SECTION NUMBER:
07	
08	
09	
10	
11	1. CANCEL 2. ENTER
12	480 1960
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	

State Transition Diagram for Registration System

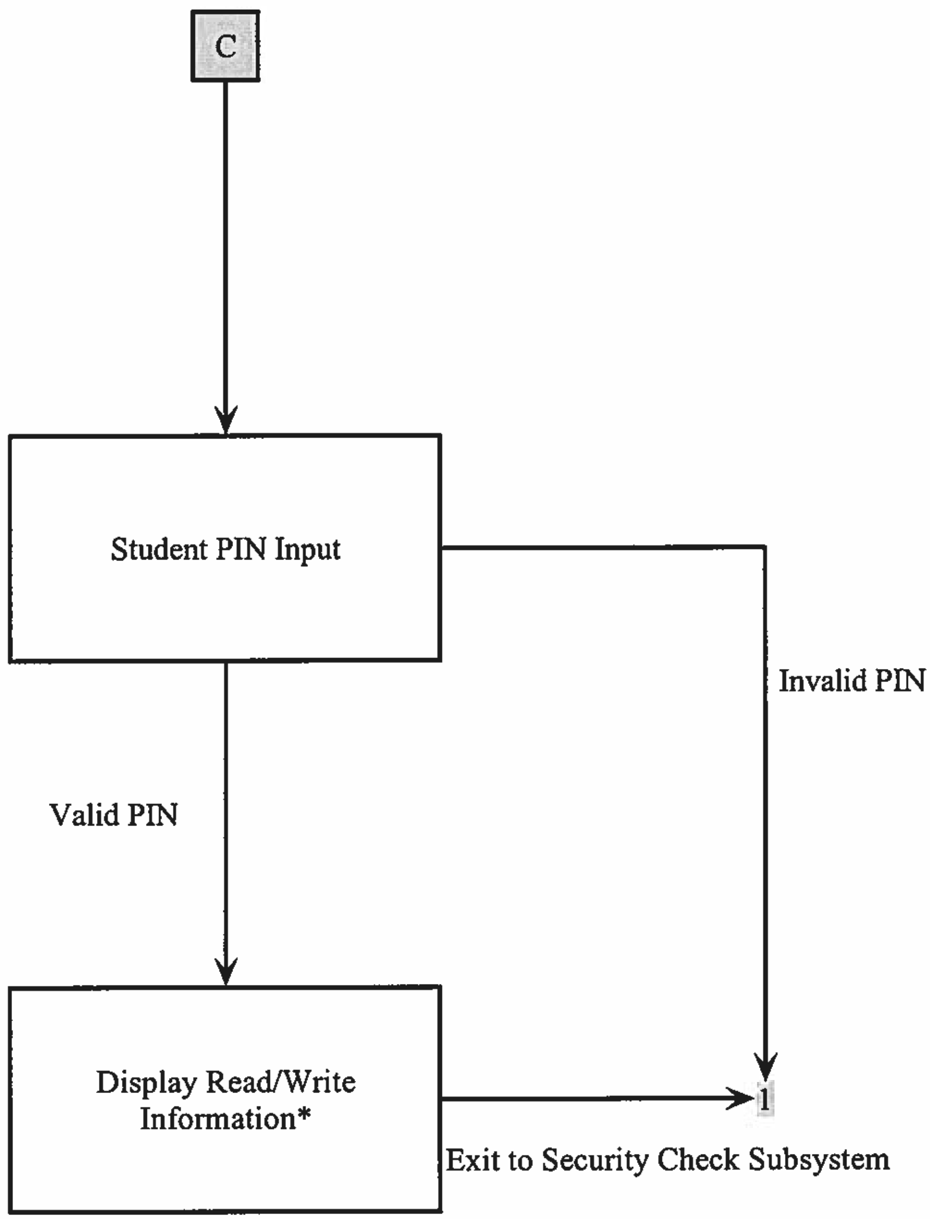


State Transition Diagram (cont...)



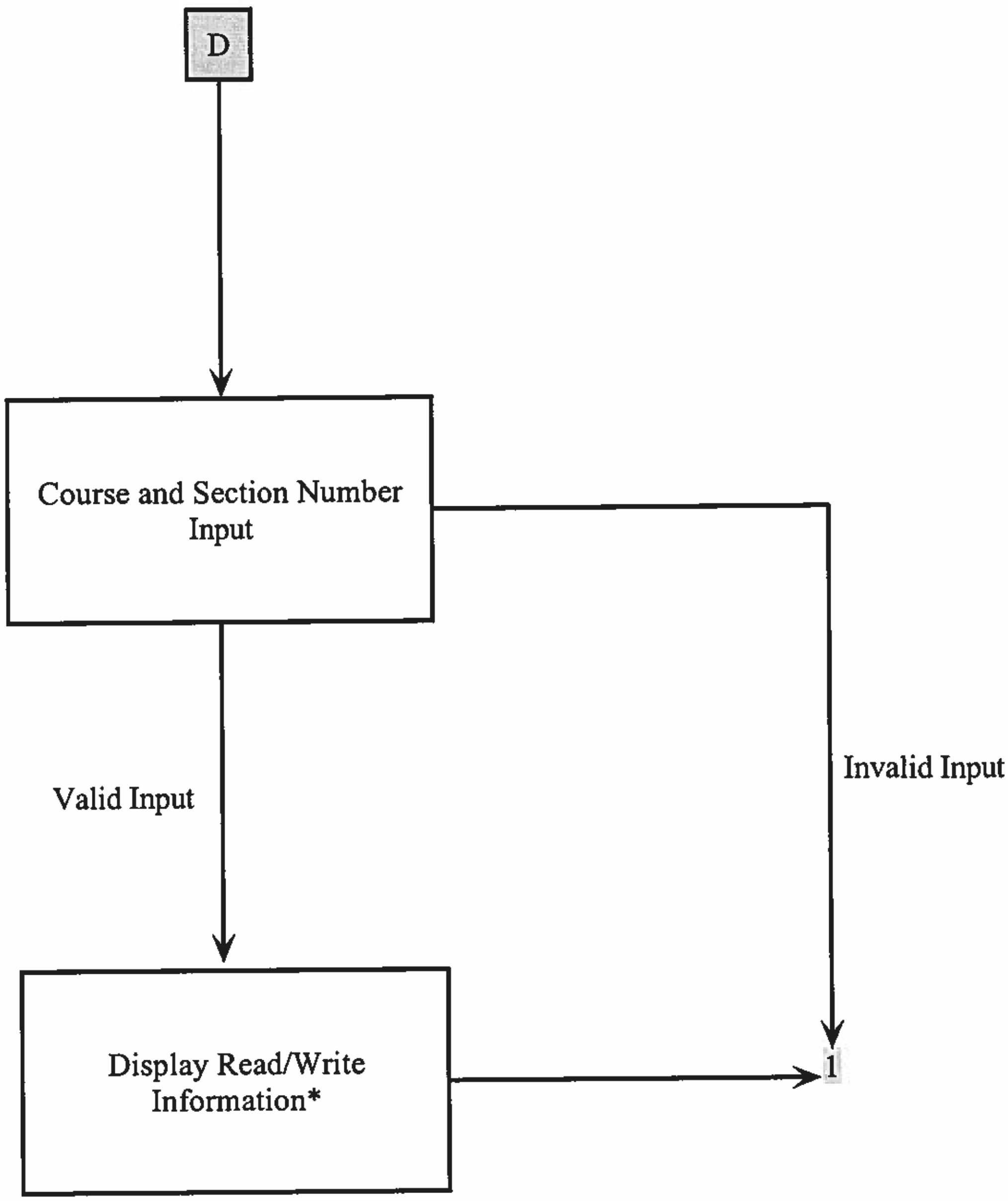
At any screen, the user has the option to click "cancel" and return to Access Validation Screen

State Transition Diagram (cont...)



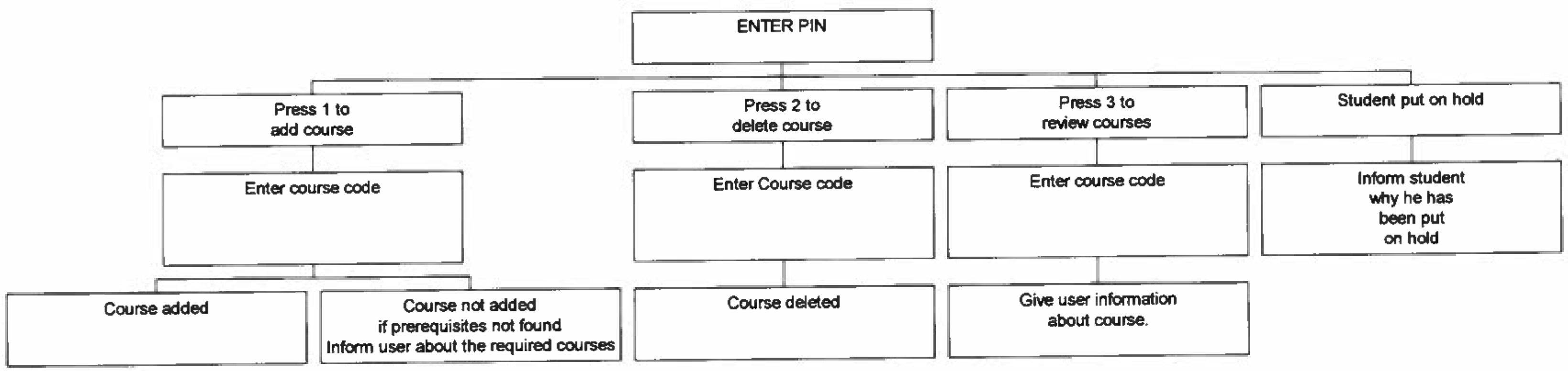
*Note: A screen is created which is in the form of the terminal Display Screen Layout Chart
At any screen, the user has the option to click "cancel" and return to Access Validation Screen

State Transition Diagram (cont...)



*Note: A screen is created which is in the form of the terminal Display Screen Layout Chart
At any screen, the user has the option to click "cancel" and return to Access Validation Screen

Phone interface for the automatic registration system



MILESTONE 13
SUBMISSION CHECKLIST AND COVER SHEET

DATE: December 8, 1997

TO: Dr. Linda Behrens

FROM: Team Members Washington, Angela
Raghuraman, Ananth

We are submitting the following milestone for your evaluation.

Milestone 13 Checklist:

[ ] We are resubmitting our System Profile from Milestone 2.

[ ] We are submitting our data flow diagrams that contain programs we packaged and designed. We circled programs appearing on the data flow diagrams to indicate which programs we designed and packaged.

[ ] We are submitting a description of the modular structure of each program. Specify the technique and tool used:
Technique:______________________________
Tool:______________________________

[ ] We are submitting a description of processing requirements for each module in our programs.

[ ] We are submitting the complete set of design specification, organized by program.

[ ] We are submitting the following additional documentation:______________________________

We have carefully checked our program specification for the following:

[X] We designed and packaged the specifications for each program appearing on our data flow diagrams (or those programs specifically assigned to us by the instructor).

[X] We factored each of our programs into modules and described the overall modular structure of the program using the tools and techniques outlined by the instructor.

[X] We described the processing logic of each program's modules.

[X] Our program specification have been packaged to clearly communicate the programming requirement to a computer programmer.

[ ] Special comments to the instructor:____________________

____

To be completed by the instructor:

I have reviewed Milestone 13. My evaluation follows:

[ ] Milestone returned ungraded because of________________

____

[ ] Milestone acceptable as submitted. Grade of_____ recorded in students' record.

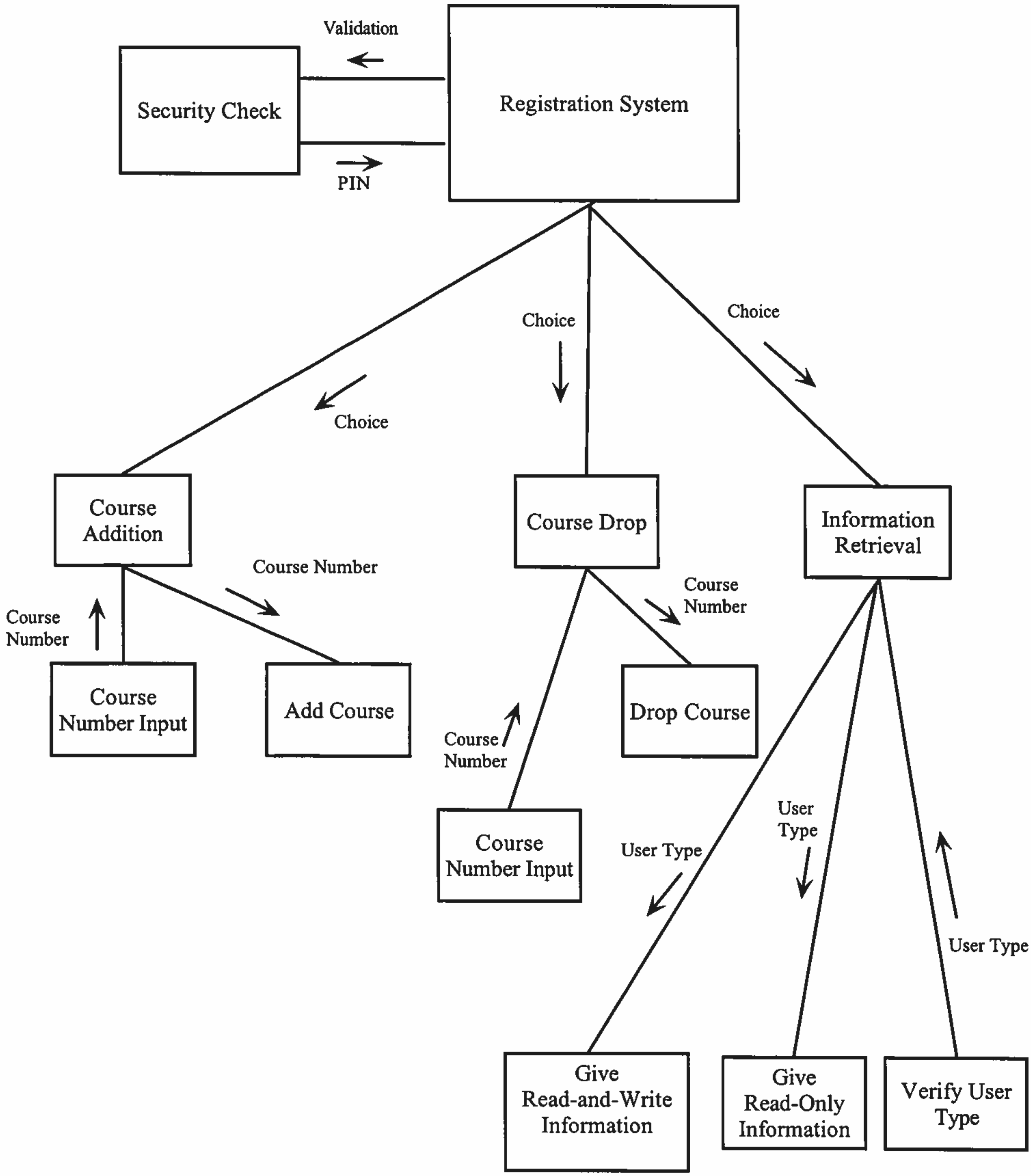
[ ] Milestone acceptable as submitted; however, it needs to be corrected and resubmitted to establish a permanent grade.

[ ] Milestone **NOT** acceptable as submitted. Grade of_____ recorded student's recorded.

[ ] Milestone **NOT** acceptable as submitted because__________

____

Resubmit by date of______________________ to establish a permanent grade.


Validation
Security Check
Registration System
PIN
Choice
Choice
Choice
Course Addition
Course Drop
Information Retrieval
Course Number
Course Number
Course Number
Course Number
Course Number Input
Add Course
Drop Course
Course Number Input
User Type
User Type
User Type
Give Read-and-Write Information
Give Read-Only Information
Verify User Type

- Registration System (1)
  - Security Check (1)
    - PIN Input
      - Data Attribute
    - PIN Verification
      - Data Attribute
    - Validate Access and allow (or not allow) user to access
  - Course Addition (N)
    - Course Number Input
      - Data Attribute
    - Add Course
  - Course Drop (1)
    - Course Number Input
      - Data Attribute
    - Drop Course
  - Information Retrieval (1)
    - Give Read-and-Write Information
    - Verify User Type
    - Give Read-Only Information

Structured English Descriptions of Modular Structure of Programs

1. Get Course Number from Student
2. Check for Validity of Course Number
3. Check if Student will have 12 credits after drop
4. If (2) and (3) are satisfied, drop course
   else don't drop course

1. Search in Student and Univeristy Records
   to determine Course Number
2. If User Type=Student then
       give Read-Only Information
   else Give Read-and-Write Information

Structured English Description of Modular Structurd of Programs

Security Check

Begin Security Check
1. Input Pin
2. Vertify Pin
If Pin Is Valid
  Then allow User To Acess

Course Add

1. Get Couse Number from Student
2. Check for Validity Course Number
3. **If** course number is valid
   then add course

Didn't follow General System Report Outline p. 133.

MILESTONE 14

SUBMISSION CHECKLIST AND COVER SHEET

MS14 tab missing

DATE: December 13, 1997

TO : Dr. Linda Behrens

FROM: TEAM MEMBERS WASHINGTON, ANGELA
RAGHURAMAN, ANANTH

We are submitting the following milestone for your evaluation

Milestone 14 checklist:

[X] We are submitting a formal report.

[X] We are submitting an informal notebook into which our system specification have been organized.

We have carefully checked our submission for the following:

[X] There are no spelling errors.

[X] There are no sentence fragments, comma splices, or run-on sentences.

[X] Sentences structure has been carefully checked for noun-verb agreement.

[X] Sentences have been carefully checked for proper punctuation.

[X] Technical terms have been eliminated from narratives intended for non-technical users.

To be completed by the instructor:

I have reviewed Milestone 14. My evaluation follows:

[ ] Milestone returned ungraded because of________________________

____

—3.

[ ] Milestone acceptable as submitted. Grade of__________ recorded in students' record.

[ ] Milestone acceptable as submitted; however, it needs to be corrected and resubmitted to establish a permanent grade.

Incomplete late!

[✓] Milestone **NOT** acceptable as submitted. Grade of 20 recorded in students' record.

[ ] Milestone **NOT** acceptable as submitted because ____________

____

Resubmit by date of____________________ to establish a permanent grade.

AUTOMATICA INC.
9213, UNIVERSITY BOULEVARD
CHARLESTON, SC, 29423
803-859-9277
Fax#: 803-859-9123

December 13, 1997

Rex Nestor
Head Registrar
CSU
9200 University Boulevard
Charleston, SC 29423

ss

Dear Mr Nestor :

&

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We have copleted the systems analysis and design for your new registration system and are glad to present you with the final technical report on it. This report summarizes and documents the systems development process.

ss

Sincerely,

Ananth Raghuraman, Angela Washington
Systems Analysts

Introduction

Purpose

The purpose of this report is to appraise you of the method used in building the registration system and to document it. It gives a full picture of the actual development process. In keeping with the principle that owners and users should be involved, we feel ~~that~~ this report will show what we accomplished and let you ~~to~~ decide whether the project should be revised or whether it is satisfactory. We would appreciate if you reviewed it and present us with your opinion.

Background

You may remember that you had initially asked us to evaluate and decide on a new registration system, which would enhance your productivity; accordingly, we performed a feasibility analysis and determined that the original system was inefficient and resulted in low morale among students, whose views we were prompt to consider. We found a solution in automating the system. This, we hope, would make the system more efficient, by allowing students to register themselves without the hassle of registrars and manual errors.

Scope

The new system involves automating the existing registration system. This is done by allowing students to register via a touch-tone phone. space The students, are now saved the bother of going to the registrar's office to register and also assured of minimum error. Other users who benefit from the system are the assistant registrars.

Report Stucture

We are presenting our report in a factual format, which we hope, you would find useful. This format can be described as follows:

- I Introduction
- II Methods and procedures
- III Facts and details
- IV Discussion and analysis of facts
- V Recommendations
- VI Conclusion

Specifications

We are including legends and hardware and software constraints.

System Overview

The DFDs for the programmer and user are enclosed. The first one is for you and other managers, Mr. Nestor, and the other is for the use of any programmer(s) you might hire in the future.

Database Specifications

The structure charts, terminal dialogue charts, and terminal screen display layout charts are enclosed.

Project repository entries for the main programs are also enclosed.

Analysis of the system

* System performance: The throughput (total number of students processed) and response time (time taken to process each student) of the existing system is unsatisfactory. There have been long delays due to incorrect data entry,

unavailability of course ststus (whether a course was availabile or not), which have resulted in decreased performance.

* Data and information:The data regarding availability of courses, at the time of registration, is obsolete. Many courses might already have been taken and the student is left high and dry as to which course he should take. Also, the manual data entry into the existing computer system often results in data entry errors which result in frustration to students.

Implementation Schedule

We are enclosing a GANTT chart to give you an idea of what the time schedule associated with the project would be.

Recommendations and conclusions

We strongly feel that our proposal to automate your registration system will overcome the present problems with the system. Also we recommed that all users and managers such as yourself, be involved throughout the development of the system.

Expanded Project Repository Entries for Outputs

TYPE:	STUDENT REPORT
EXPLODES TO:	REPORT ON COURSE DROP (page 2)
	REPORT ON COURSE ADD (page 3)
	INFORMATION RETRIEVAL (page 4)
DESCRIPTION:	THIS REPORT IS GENERATED WHEN THE STUDENT ATTEMPTS TO ADD OR DROP COURSES OR WHEN HE/SHE GETS INFORMATION ON AVAILABLE COURSES.
MEDIUM:	PHONE INTERFACE
OUTPUT CHARACTERISTICS:	IT IS A DATA FLOW FROM THE COURSE ADDITION, COURSE DROP AND INFORMATION SUBSYSTEMS. AS TO INFORMATION RETRIEVAL, THE STUDENT CAN ACCESS READ-ONLY INFORMATION CONCERNING AVAILABLE COURSES AND HIS/HER OWN RECORDS. THE REPORT IS VOCAL, COMMUNICATED BY A RECORDED VOICE.
OUTPUT COMPOSITION:	COURSE NUMBERS
	REASONS FOR SUCCESS IN ATTEMPT OR LACK OF IT

TYPE: REPORT ON COURSE DROP

DESCRIPTION: THIS REPORT IS GENERATED WHEN THE STUDENT ATTEMPTS TO DROP COURSES. IT TELLS THE STUDENT WHETHER A COURSE WAS DROPPED OR NOT AND THE REASONS FOR NOT DROPPING A COURSE.

MEDIUM: PHONE INTERFACE

OUTPUT CHARACTERISTICS: THIS DATA FLOWS FROM COURSE DROP SUBSYSTEM TO STUDENT. REPORTS ARE IN THE FORM OF A RECORDED VOICE WHICH IS GENERATED IN THE COMPUTER CENTER OF THE UNIVERSITY. REPORTS OCCURS WITH EVERY INSTANCE WHERE A STUDENT TRIES TO DROP A COURSE.

OUTPUT COMPOSITION: COURSE NUMBERS OF COURSES DROPPED

REASONS FOR NOT DROPPING A COURSE

TYPE:	REPORT ON COURSE ADD
DESCRIPTION:	THIS REPORT IS GENERATED WHEN A STUDENT ATTEMPTS TO ADD COURSES. IT TELLS THE STUDENT WHETHER A COURSE WAS ADDED OR NOT AND THE REASONS FOR NOT ADDING A COURSE.
MEDIUM:	PHONE INTERFACE
OUTPUT CHARACTERISTICS:	THIS DATA FLOWS FROM COURSE ADDTION SUBSYSTEM TO STUDENT. THE REPORTS ARE IN THE FORM OF A RECORDED VOICE THAT'S GENERATED IN THE COMPUTER CENTER OF THE UNIVERSITY WHICH OCCURS WITH EVERY INSTANCE A STUDENT ATTEMPTS TO ADD A COURSE.
OUTPUT COMPOSITION:	COURSE NUMBERS REASONS FOR ADDING/NOT ADDING A COURSE

TYPE:	INFORMATION RETRIEVAL
EXPLODES TO:	READ-AND-WRITE INFORMATION REPORT (page 5)
	READ-ONLY INFORMATION REPORT (page 6)
MEDIUM:	PHONE INTERFACE AND COMPUTER TERMINALS

OUTPUT

CHARACTERISTICS: IT IS A DATA FLOW FROM INFORMATION RETRIEVAL SUBSYSTEM TO THE STUDENT. IT IS GENERATED IN THE FORM OF A RECORDED VOICE, IN CASE A STUDENT IS USING A PHONE AND COMPUTER SCREEN IN CASE A UNIVERSITY AUTHORITY IS USING A TERMINAL.

OUTPUT

COMPOSITION: COURSE NUMBERS OF AVAILABLE COURSES, COURSE NUMBERS OF UPDATED COURSES, SOCIAL SECURITY NUMBERS OF STUDENTS WHOSE RECORD HAS BEEN UPDATED.

TYPE:	READ-AND-WRITE INFORMATION
EXPLODES TO:	DETAIL REPORT (page 7)
	SUMMARY REPORT (page 8)
	EXCEPTION REPORT (page 9)
MEDIUM:	COMPUTER TERMINAL
OUTPUT CHARACTERISTICS:	THIS DATA FLOWS FROM THE INFORMATION RETRIEVAL SUBSYSTEM TO STUDENT. THE REPORTS ARE IN THE FORM OF COMPUTER SCREENS THAT IS GENERATED IN THE COMPUTER CENTER. THE REPORT OCCURS WITH EVERY INSTANCE IN WHICH A UNIVERSITY AUTHORITY ATTEMPTS TO RETRIEVE REPORTS OR STUDENT RECORDS.
OUTPUT COMPOSITION:	STUDENT RECORDS

TYPE:	READ-ONLY-INFORMATION REPORT
DESCRIPTION:	THIS REPORT IS GENERATED WHEN A STUDENT ACCESSES THE SYSTEM TO GET INFORMATION. IT GIVES HIM/HER THE LIST OF AVAILABLE COURSES, INFORMATION ON HIS/HER OWN RECORDS.
MEDIUM:	PHONE INTERFACE
OUTPUT CHARACTERISTICS:	THIS DATA FLOWS FROM INFORMATION RETRIEVAL SUBSYSTEM TO THE STUDENT. THE REPORTS ARE IN THE FORM OF A RECORDED VOICE. THE REPORTS ARE GENERATED IN THE COMPUTER CENTER OF THE UNIVERSITY, AND IT OCCURS WITH EVERY INSTANCE WHERE A STUDENT ACCESS THE SYSTEM TO RETRIEVE INFORMATION.
OUTPUT COMPOSITION:	COURSE NUMBERS OF AVAILABLE COURSES, INFORMATION ON HIS/HER OWN RECORD.

TYPE:	DETAIL REPORT
DESCRIPTION:	THIS REPORT GIVES THE UNIVERSITY AUTHORITY, REGISTRATION INFORMATION ON ALL STUDENTS AND COURSES.
MEDIUM:	COMPUTER TERMINAL
OUTPUT CHARACTERISTIC:	THIS DATA FLOW FROM INFORMATION RETRIEVAL SUBSYSTEM TO STUDENT. DETAIL REPORT IS IN THE FORM OF COMPUTER SCREEN THAT IS GENERATED IN THE COMPUTER CENTER.
OUTPUT COMPOSITION:	STUDENT RECORDS COURSE LIST, PREREQUISITE LIST

TYPE:	SUMMARY REPORT
DESCRIPTION:	THIS REPORT SUMMARIZES THE REGISTRATION INFORMATION OF EACH STUDENT AND GIVES ONLY HIS/HER MAJOR, CREDITS EARNED, GPA AND TUITION STATUS (WHETHER THE STUDENT PAID HIS/HER TUITION OR NOT).
MEDIUM:	COMPUTER TERMINAL
OUTPUT CHARACTERISTICS:	IT IS A DATA FLOW FROM INFORMATION RETRIEVAL SUBSYSTEM TO UNIVERSITY AUTHORITY. IT IS IN THE FORM OF COMPUTER SCREENS.
OUTPUT COMPOSITION:	STUDENT RECORDS

Expanded Project Repository Entries for Inputs

TYPE:	PIN
DESCRIPTION:	IT IS A NUMBER GIVEN TO A STUDENT WHICH UNIQUELY INDENTIFIES A USER AS WELL AS DECIDES WHAT TYPE OF INFORMATION HE/SHE MAY ACCESS.
MEDIUM:	PHONE INTERFACE/COMPUTER TERMINAL
INPUT CHARACTERISTICS:	IT IS A DATA FLOW FROM THE STUDENT TO THE REGISTRATION SYSTEM.
INPUT COMPOSITION:	FOUR-DIGIT ALPHA-NUMERIC STRING

TYPE:	EXCEPTION REPORT
DESCRIPTION:	DEPENDING UPON THE CRITERIA, THIS GIVES ALL THE STUDENT RECORDS WHICH SATISFY THE CRITERIA.
MEDIUM:	COMPUTER TERMINAL
OUTPUT CHARACTERISTICS:	IT IS A DATA FLOW FROM INFORMATION RETRIEVAL SUBSYSTEM TO UNIVERSITY AUTHORITY. IT IS GENERATED IN THE COMPUTER CENTER.
OUTPUT COMPOSITION:	STUDENT RECORDS

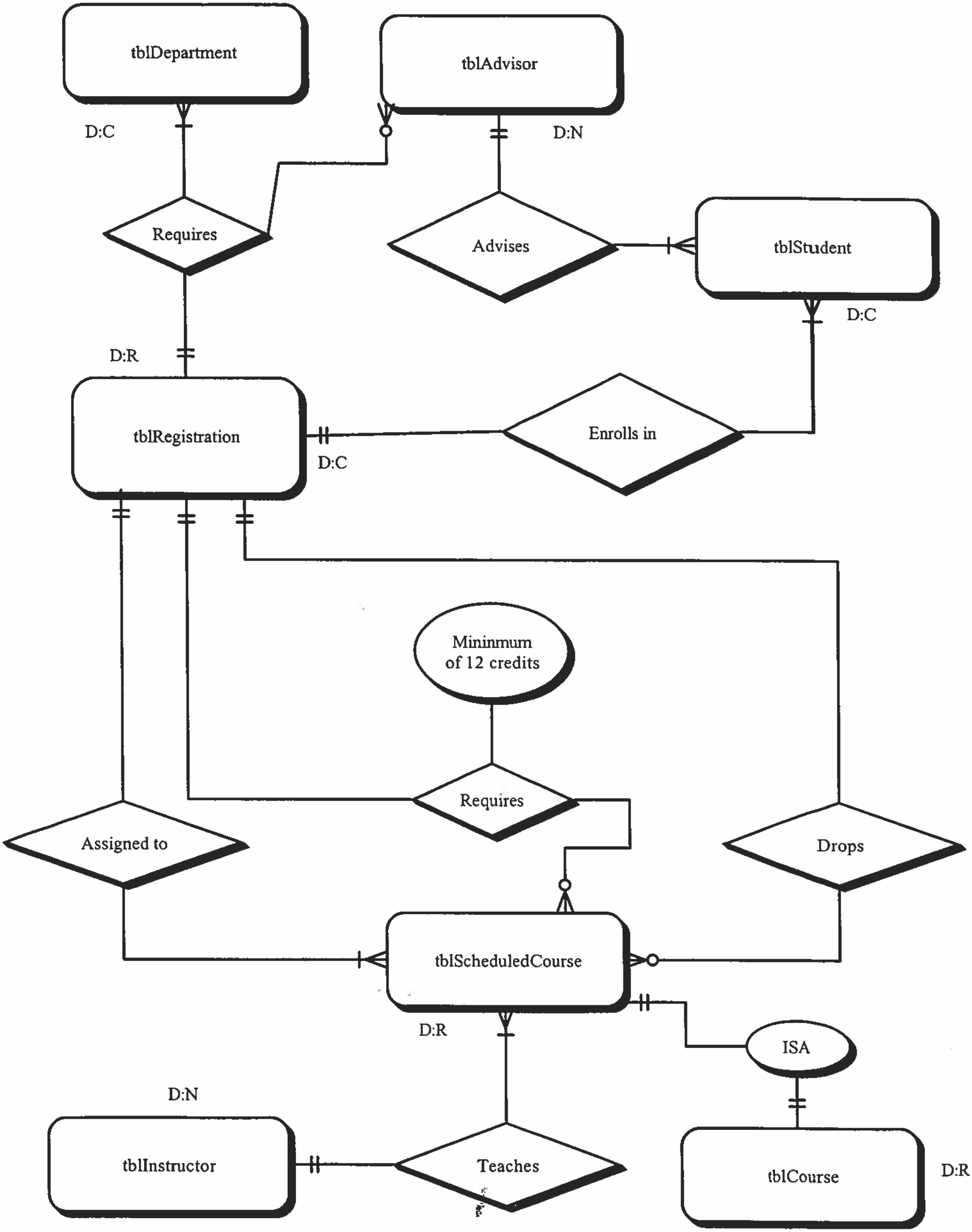
Project Repository Entries for Inputs (cont...)

TYPE:	COURSE NUMBER
DESCRIPTION:	A COURSE CHARACTEIRZED BY ITS COURSE NUMBER AND THE ACTUAL INPUT IS THE COURSE NUMBER. THE COURSE NUMBER UNIQUELY IDENTIFIES A COURSE. IT IS INPUT WHEN THE USER WANTS DETAILS OF A COURSE.
MEDIUM:	PHONE INTERFACE/COMPUTER TERMINAL
INPUT CHARACTERISTICS:	COURSE NUMBER IS A DATA FLOW FROM THE USER TO THE INFORMATION RETRIEVAL SUBSYSTEM OF THE REGISTRATION SYSTEM. A USER CAN SEARCH FOR DETAILS OF A COURSE USING THE TWO DIGIT COURSE NUMBER.
INPUT COMPOSITION:	TWO DIGIT NUMBER

TYPE:	COURSE NUMBER, SECTION NUMBER
DESCRIPTION:	IT IS A NUMERIC STRING WHICH UNIQUELY IDENTIFIES A COURSE. THE USER CAN SEARCH FOR THE DETAILS OF A COURSE USING THE COURSE NUMBER. SEARCHING USING THE COURSE NUMBER GIVES DETAILS OF ALL SECTIONS OF THE COURSE
INPUT COMPOSITION:	2 DIGIT NUMERIC STRING

TYPE:	SECTION NUMBER
DESCRIPTION:	IT IS A NUMERIC STRING WHICH WHEN COMBINED WITH COURSE NUMBER, UNIQUELY IDENTIFIES A SECTION OF A COURSE.
MEDIUM:	PHONE INTERFACE/COMPUTER TERMINAL
INPUT CHARACTERISTICS:	IT IS AN INPUT FROM THE USER TO THE SYSTEM
INPUT COMPOSITION:	2 DIGIT NUMERIC STRING

Physical Database Schema
Note: All attributes are listed on the next page due to paucity of space



Attributes of Entities

(Physical Database Schema cont...)

tblDepartment		
colDepartmentNumber	varchar(2)	[PK]
colDepartmentName	varchar(10)	

tblAdvisor		
colAdvisorLastName	varchar(30)	[PK1]
colAdvisorSocial	varchar(11)	[PK2]
colAdvisorFirstName	varchar(15)	

Attributes of Entities cont…

tblStudent Physical Display		
colStudentPIN	varchar(4)	[PK1]
colStudentSocial	varchar(11)	[PK2]
colStudentLastName	varchar(30)	
colStudentFirstName	varchar(15)	
colStudentType	varchar(4)	
colTerm	varchar(5)	
colRace	varchar(10)	
colMajor	varchar(9)	
colMinor	varchar(9)	
colSex	char(1)	
colDOB	varchar(8)	
colTempAdd	varchar(40)	
colPermAdd	varchar(40)	
colPhNumber	varchar(10)	
colCrEarned	int	
colVAStatus	int	
colDayStudent	int	
colDegreeStatus	int	
colCSUAthlete	varchar(15)	

Attributes of Entities cont...

tblCourse Physical Display		
colCourseNumber	varchar(2)	[PK]
colCourseTitle	varchar(10)	
colNumberOfCr	int	
colDeptNumber	varchar(2)	[FK]

tblRegistration Physical Display		
colCourseNumber	varchar(2)	[PK] [FK]
colSectionNumber	varchar(2)	[PK] [FK1]
colStudentSocial	varchar(11)	[PK] [FK2]

Note: Please see next page for relationships.

The table Registration is related to:

1. table Student by key StudentSocial
2. table Course by key CourseNumber

Attributes of Entities cont...

tblInstructor Physical Display		
colInstSocial	varchar(11)	[PK]
colInstLastName	varchar(30)	[PK1]
colInstFirstName	varchar(15)	
colInstDegree	varchar(30)	
colIstOffice	varchar(40)	
colInstPhNumber	varchar(10)	
colDeptNumber	varchar(2)	[FK]

Attributes of Entities cont…

tblScheduledCourse Physical Display		
colSectionNumber	varchar(2)	[PK1]
colCourseNumber	varchar(2)	[PK2] [FK]
colRoomNumber	varchar(2)	
colBuildingNumber	varchar(2)	
colStartTime	varchar(5)	
colEndTime	varchar(5)	
colDaysSch	varchar(14)	
colInstLastName	varchar(30)	[FK1]
colInstFirstName	varchar(15)	

The above table is related to the table Instructor by the foreign key, InstLastName and to the table Course by foreign key CourseNumber.

3. table Scheduled Course by key SectionNumber

The table Course is related to table Department by foreign key DepartmentNumber

RECORD LAYOUT CHART FOR REGISTRATION FILE
name of file

Field Name	COURSE NUMBER	SECTION NUMBER
Characteristics*	ALPHA STRING	ALPHA STRING
Position**	0 1	3 4

STUDENT SOCIAL	
ALPHA STRING	
5 15	

**POSITION
Hexadecimal / Decimal
Numbering
from
00 to FF / 0 to 255

HEXADECIMAL	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	10	11	12	13	14	15	16
DECIMAL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
HEX	40	41	42	43	44	45	46	47	48	49	4A	4B	4C	4D	4E	4F	50	51	52	53	54	55	56
DEC	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
HEX	80	81	82	83	84	85	86	87	88	89	8A	8B	8C	8D	8E	8F	90	91	92	93	94	95	96
DEC	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
HEX	C0	C1	C2	C3	C4	C5	C6	C7	C8	C9	CA	CB	CC	CD	CE	CF	D0	D1	D2	D3	D4	D5	D6
DEC	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214

RECORD LAYOUT CHART FOR COURSE FILE
name of file

Field Name	COURSE NUMBER	TITLE	CREDITS
Characteristics*	ALPHA STRING	ALPHA STRING	INTEGER
Position**	0 1	12 21	23 23

DEPARTMENT NUMBER	
ALPHA STRING	
25 26	

**POSITION
Hexadecimal / Decimal
Numbering
from
00 to FF / 0 to 255

HEXADECIMAL	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	10	11	12	13	14	15	16
DECIMAL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
HEX	40	41	42	43	44	45	46	47	48	49	4A	4B	4C	4D	4E	4F	50	51	52	53	54	55	56
DEC	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
HEX	80	81	82	83	84	85	86	87	88	89	8A	8B	8C	8D	8E	8F	90	91	92	93	94	95	96
DEC	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
HEX	C0	C1	C2	C3	C4	C5	C6	C7	C8	C9	CA	CB	CC	CD	CE	CF	D0	D1	D2	D3	D4	D5	D6
DEC	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214

RECORD LAYOUT CHART FOR STUDENT FILE
name of file

Field Name	Characteristics*	Position** (from)	Position** (to)
STUDENT PIN	NUMERIC STRING	0	3
STUDENT SOCIAL	NUMERIC STRING	4	14
STUDENT LAST NAM	ALPHA STRING	15	4
STUDENT FIRST NAME	ALPHA STRING	46	65
STUDENT TYPE	ALPHA STRING	67	70
TERM	ALPHA NUMERIC STRING	72	75
RACE	ALPHA STRING	77	86
MAJOR	ALPHA STRING	88	96
MINOR	ALPHA STRING	98	106
SEX	CHARACTER	108	10
DOB	ALPHA STRING	109	116
TEMP ADD	ALPHA STRING	118	157
PERM ADD	ALPHA STRING	159	198
Ph NUMBER	ALPHA STRING	199	2
CREDITS EARNED	~~209~~ INTEGER Ø	209	210
VA STATUS	INTEGER	211	211
DAY STUDENT	INTEGER	213	213
DEGREE STATUS	INTEGER	215	215
CSU ATHLETE	ALPHA STRING	217	231

**POSITION
Hexadecimal / Decimal
Numbering
from
00 to FF / 0 to 255

HEXADECIMAL	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	10	11	12	13	14	15	16	1
DECIMAL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	2
HEX	40	41	42	43	44	45	46	47	48	49	4A	4B	4C	4D	4E	4F	50	51	52	53	54	55	56	5
DEC	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	8
HEX	80	81	82	83	84	85	86	87	88	89	8A	8B	8C	8D	8E	8F	90	91	92	93	94	95	96	9
DEC	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	15
HEX	C0	C1	C2	C3	C4	C5	C6	C7	C8	C9	CA	CB	CC	CD	CE	CF	D0	D1	D2	D3	D4	D5	D6	D
DEC	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	21

RECORD LAYOUT CHART FOR DEPARTMENT FILE
name of file

Field Name	DEPARTMENT NUMBER	DEPARTMENT NAME	
Characteristics*	NUMERIC STRING	NUMERIC STRING	
Position**	0 1	11 20	

**POSITION
Hexadecimal / Decimal
Numbering
from
00 to FF / 0 to 255

HEXADECIMAL	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	10	11	12	13	14	15	16	17
DECIMAL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
HEX	40	41	42	43	44	45	46	47	48	49	4A	4B	4C	4D	4E	4F	50	51	52	53	54	55	56	57
DEC	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87
HEX	80	81	82	83	84	85	86	87	88	89	8A	8B	8C	8D	8E	8F	90	91	92	93	94	95	96	97
DEC	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151
HEX	C0	C1	C2	C3	C4	C5	C6	C7	C8	C9	CA	CB	CC	CD	CE	CF	D0	D1	D2	D3	D4	D5	D6	D7
DEC	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215